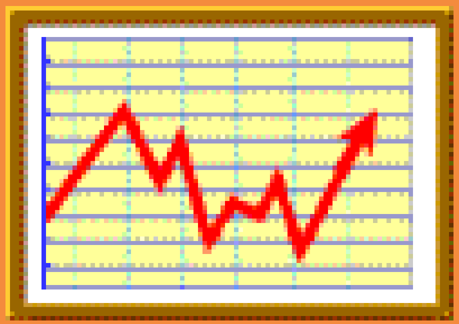
California State University, East Bay
College of Business and Economics

BAN 673 Time Series Analytics

Autocorrelation and Autoregressive Models

Lecture Materials



Dr. Z. Radovilsky

Lecture Objectives

- **Apply autocorrelation for time series and forecast residuals**
- **Identify and explain autoregressive (AR) models in time series analytics**
- **Utilize autoregressive models in time series forecasting – two-level forecasting with AR model for residuals**
- **Apply autoregressive models to identify predictability of time series**
- **Use R to develop autocorrelation and autoregressive models for time series forecasting**

Why Use Autocorrelation?

- In ***cross-sectional data***, the assumption is that a response variable and predictors don't have correlation between successive records
- In ***time series data***, values in neighboring periods tend to be correlated
  - Referred to as ***autocorrelation***
- Time series regression model can account for trend and seasonal patterns
  - However, regression model may not account for *autocorrelation*
- *Autocorrelation* is informative
  - Positive or negative, strong in certain lags (first lag or seasonal lags)
  - Contains useful information that can be used to improve forecasting

Autocorrelation

- ***Autocorrelation*** **measures how strong the values of a time series are related to their own past values**
- **Technically: compute the correlation between the series and the lagged series (approximately)**
  - ***Lag(1) autocorrelation*** **= correlation between ($y_1, y_2, \ldots, y_{t-1}$) and ($y_2, y_3, \ldots, y_t$)**
  - ***Lag(k) autocorrelation*** **= correlation between ($y_1, y_2, \ldots, y_{t-k}$) and ($y_{k+1}, y_{k+2}, \ldots, y_t$)**
- **Use of autocorrelation**
  - **Check time series forecast residuals for independence (or dependence)**
  - **Model remaining information (residuals), in particular, if residual dependency exists**
  - **Evaluate predictability, i.e., whether the series is a "random walk" or can be predictable**

Autocorrelation (cont.)

- ***Autocorrelation coefficient r_k*** **is a measure of autocorrelation between a variable itself and the same variable lagged *k* periods:**

$$r_k = \frac{\sum_{t=k+1}^{n} (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^{n} (Y_t - \bar{Y})^2}$$

where:

r_k = autocorrelation coefficient for a lag of *k* periods

$\bar{Y}$ = mean of the values of the series

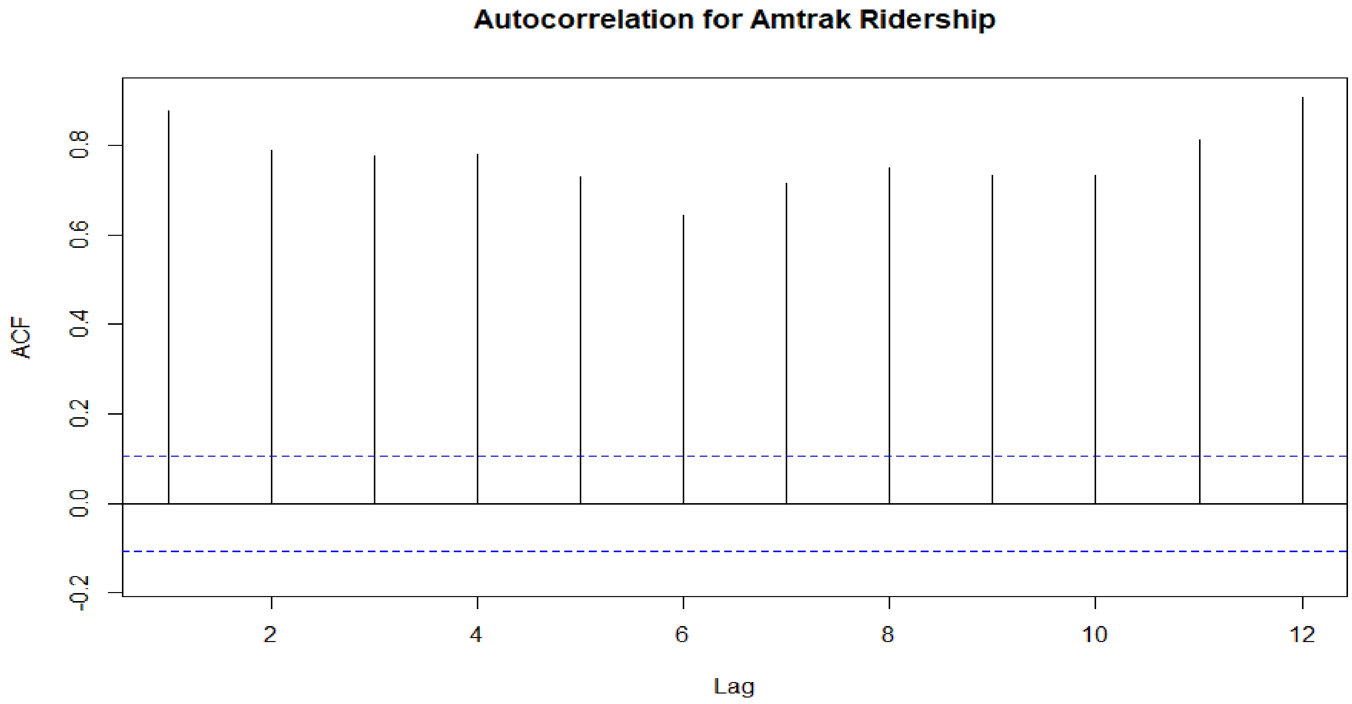
Y_t= observation in time period *t*

Y_{t-k} = observation *k* time periods earlier or at time period *t-k*

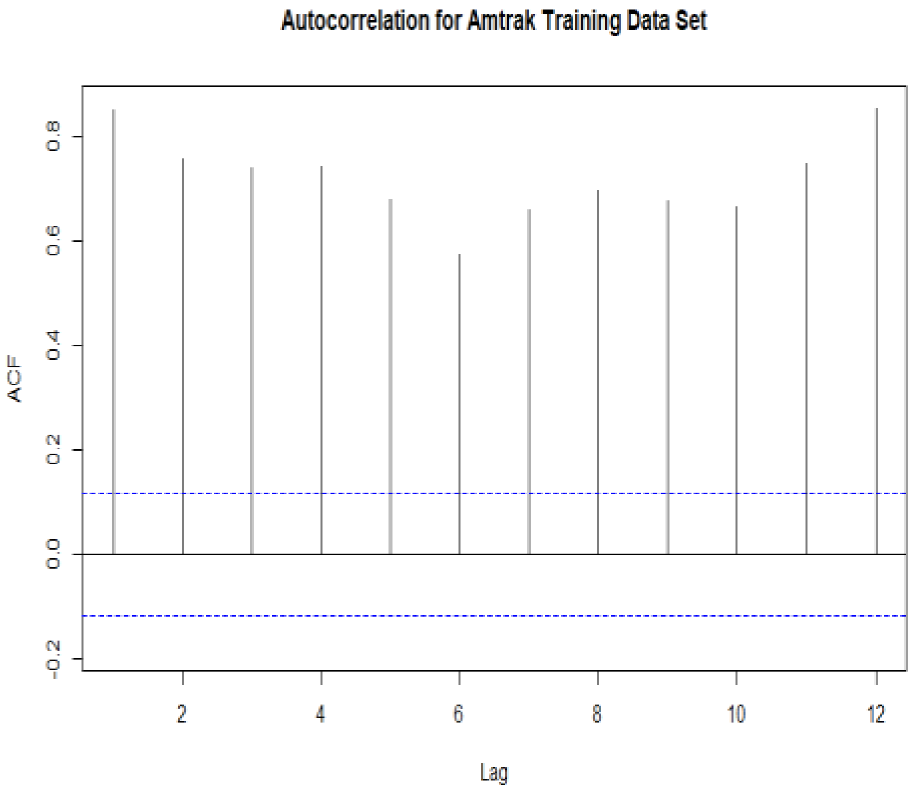
- **If a time series is *random*, the autocorrelations for any lag are close to zero**
  - **The successive values of a time series are not related to each other**

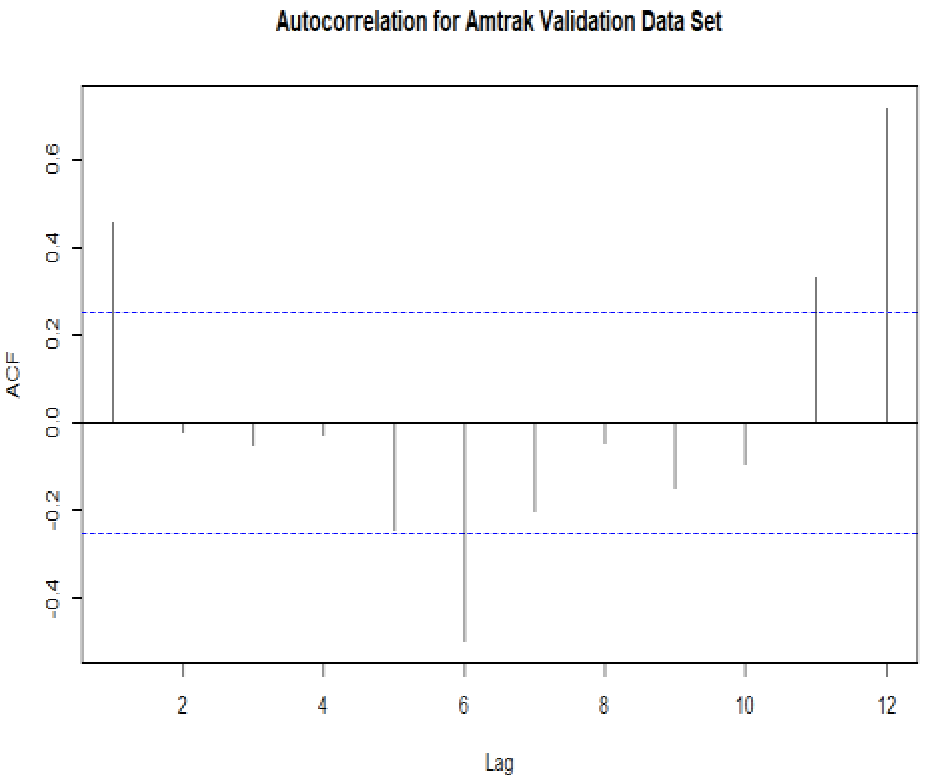
Autocorrelation Function (ACF) in R

- ***Autocorrelation function (ACF)*** **identifies and plots multiple autocorrelation coefficients for various lags of a time series data**
  - **It also calculates and plots interval (horizontal lines) to identify significant autocorrelation**



Autocorrelation for Training and Validation Data Sets





Model Relationships Identified by Autocorrelation: Autoregressive (AR) Model

- ***Autoregressive model (AR)*** **idea: model the autocorrelation directly in regression model using past observations as predictors**
- **Similar to linear regression models, the predictors are the past values of the series**
  - ***AR model of order 1, AR(1)*:**

$$Y_t = \alpha + \beta_1 Y_{t-1} + \varepsilon_t$$

  - ***AR model of order 2, AR(2)*:**

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \varepsilon_t$$

- ***Two approaches*** **in using AR models in time series forecasting**
  - ***Two-level forecasting modeling with AR model for residuals (errors)***
    - » ***Level 1*****: Use any method to generate forecasts (regression model, Holt-Winter's model, etc.)**
    - » ***Level 2*****: Examine forecast residual series for autocorrelation by utilizing time plot of forecast residuals and ACF function plot**
      - ◆ **If autocorrelation of residuals significant, fit AR model to forecast residual series**
  - ***Direct AR model***
    - » **Fit AR model directly into original series by using an *ARIMA* model**

Two-Level Forecasting with AR Model for Residuals

- **Generate *k*-step ahead forecast of the series, F_{t+k} ,using the identified forecasting method (regression model, Holt-Winter's model, etc.)**
- **Generate *k*-step ahead forecast of residuals (errors), e_{t+k} , using AR model**
  - **Typically, when autocorrelation at lag 1 exist and high, it is *sufficient to fit an AR(1) model* of the form:**

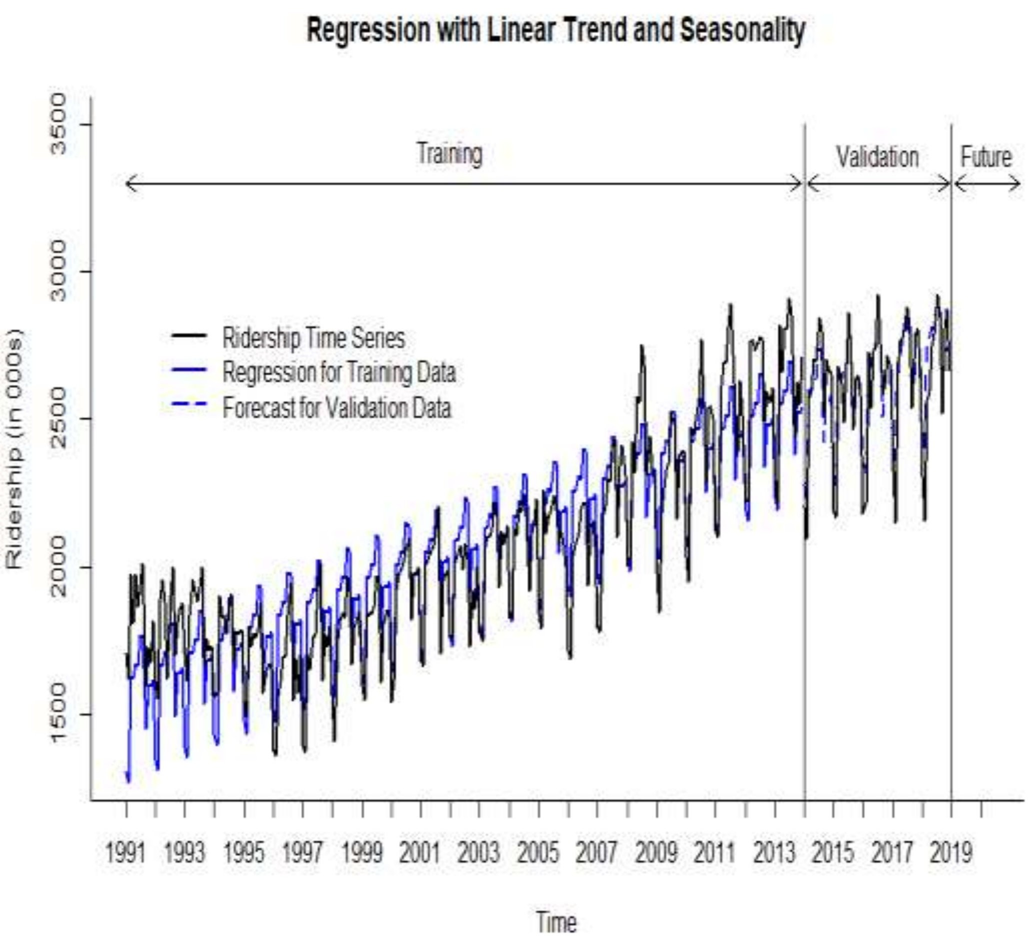
$$e_t = \alpha + \beta_1 e_{t-1} + \varepsilon_t$$

- **Improve the initial *k*-step ahead forecast of the time series by adjusting it according to its forecasted residual**
  - ***Improved forecast F^*_{t+k}* :**

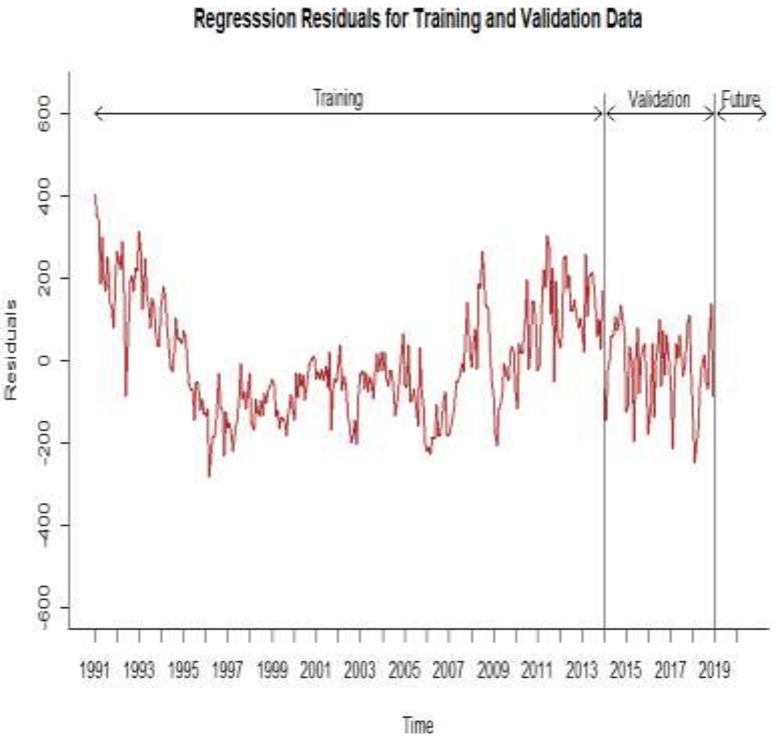
$$F^*_{t+k} = F_{t+k} + e_{t+k}$$

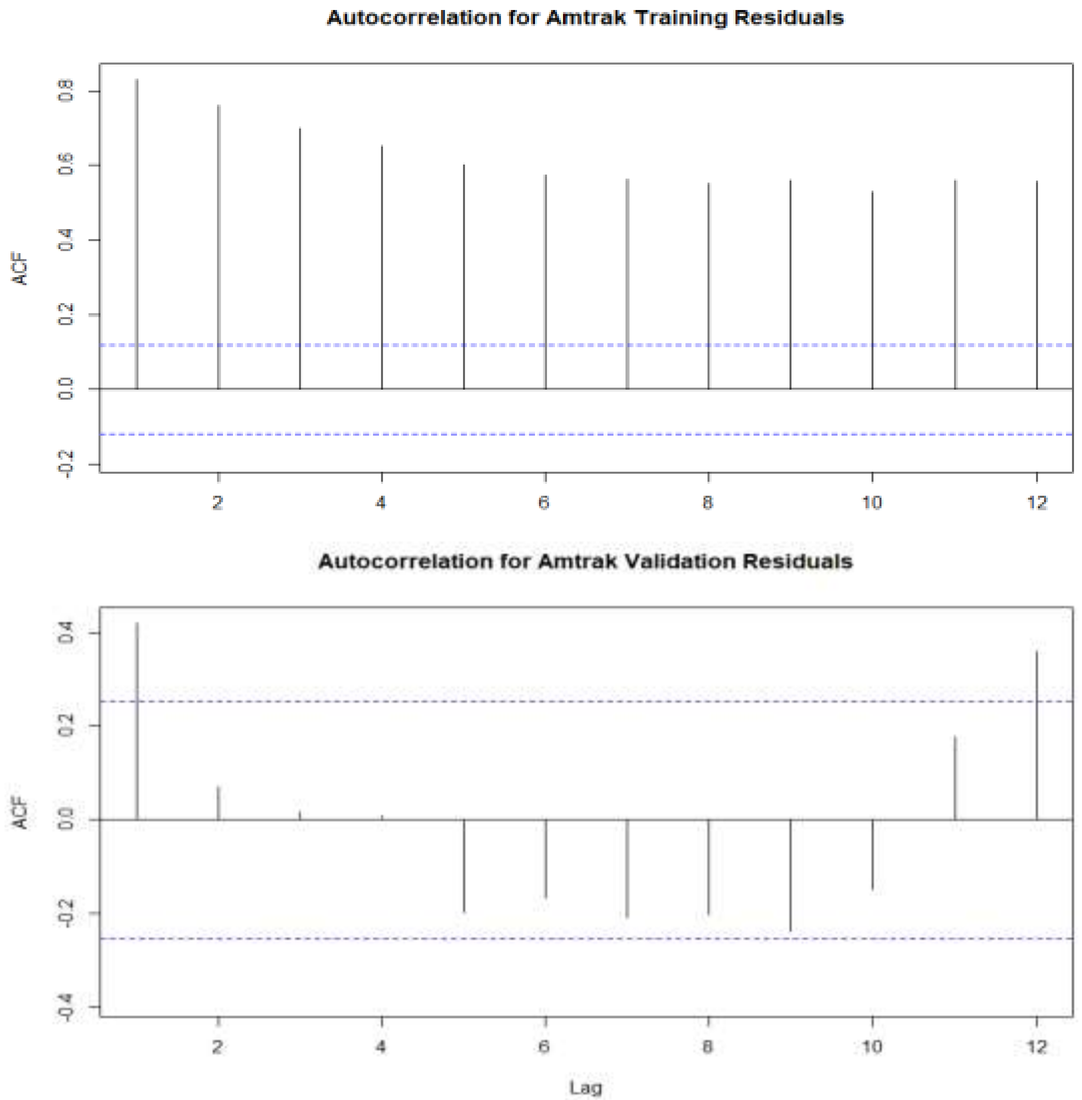
Regression with Linear Trend and Seasonality for Training Data Set

```
Call:
tslm(formula = train.ts ~ trend + season)

Residuals:
    Min      1Q  Median      3Q     Max
-281.26  -94.77  -28.53   94.12  402.96

Coefficients:
            Estimate Std.  Error    t value   Pr(>|t|)
(Intercept) 1302.4480  32.2037  40.444   < 2e-16 ***
trend          3.5132   0.1052  33.398   < 2e-16 ***
season2      -41.0741  41.0206  -1.001   0.31760
season3      316.0566  41.0210   7.705   2.68e-13 ***
season4      307.1209  41.0217   7.487   1.07e-12 ***
season5      352.9515  41.0226   8.604   7.10e-16 ***
season6      342.0432  41.0238   8.338   4.28e-15 ***
season7      441.9060  41.0253  10.772   < 2e-16 ***
season8      430.5533  41.0271  10.494   < 2e-16 ***
season9      120.0197  41.0291   2.925   0.00374 **
season10     260.0409  41.0314   6.338   1.01e-09 ***
season11     254.7954  41.0340   6.209   2.06e-09 ***
season12     269.6614  41.0368   6.571   2.66e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 139.1 on 263 degrees of freedom
Multiple R-squared:  0.8462,  Adjusted R-squared:  0.8391
F-statistic: 120.5 on 12 and 263 DF,  p-value: < 2.2e-16
```



Examine Residuals of Regression Model with Trend and Seasonality





Autoregressive AR(1) Model for Training Residuals

AR(1) Model for Residuals

Series: train.lin.season$residuals
ARIMA(1,0,0) with non-zero mean

Coefficients:

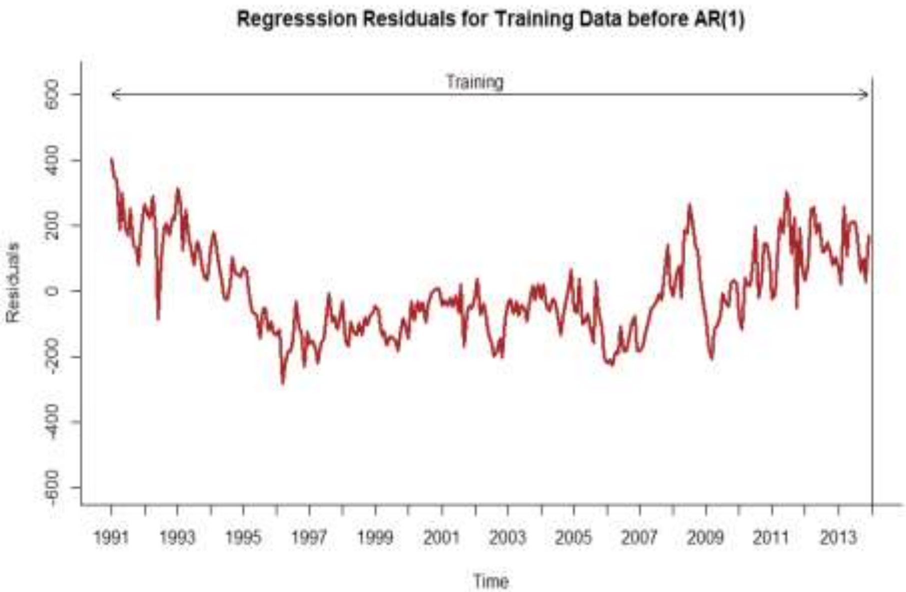
	ar1	mean
	0.8592	12.0641
s.e.	0.0322	30.4346

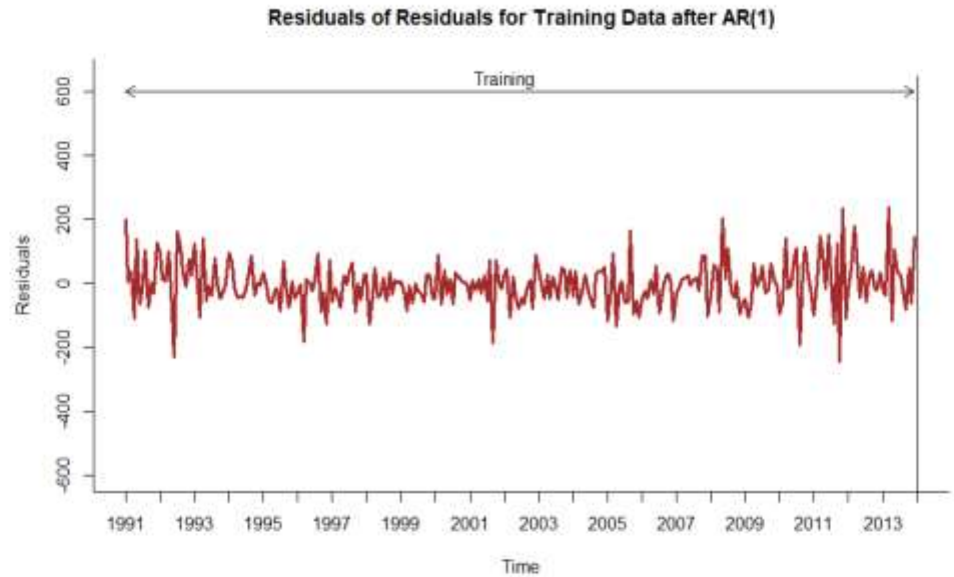
sigma^2 = 5273: log likelihood = -1574.01
AIC=3154.01 AICc=3154.1 BIC=3164.87

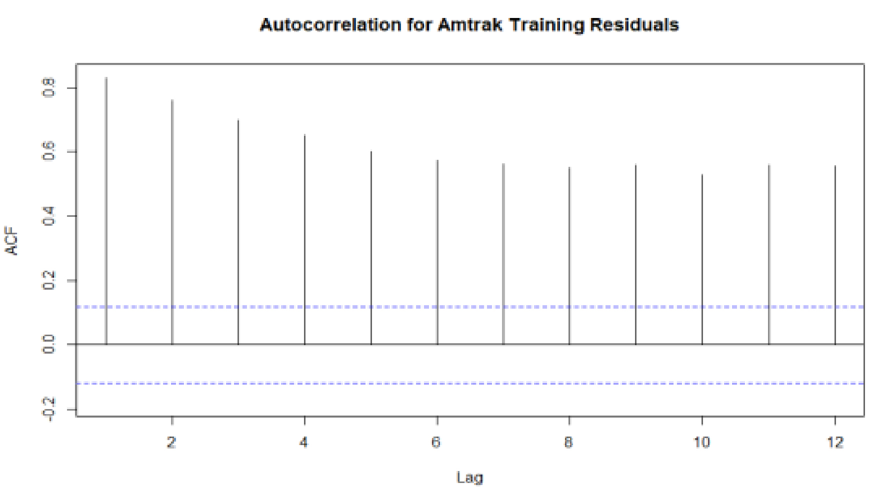
$e_t = 12.064 + 0.859 e_{t-1}$

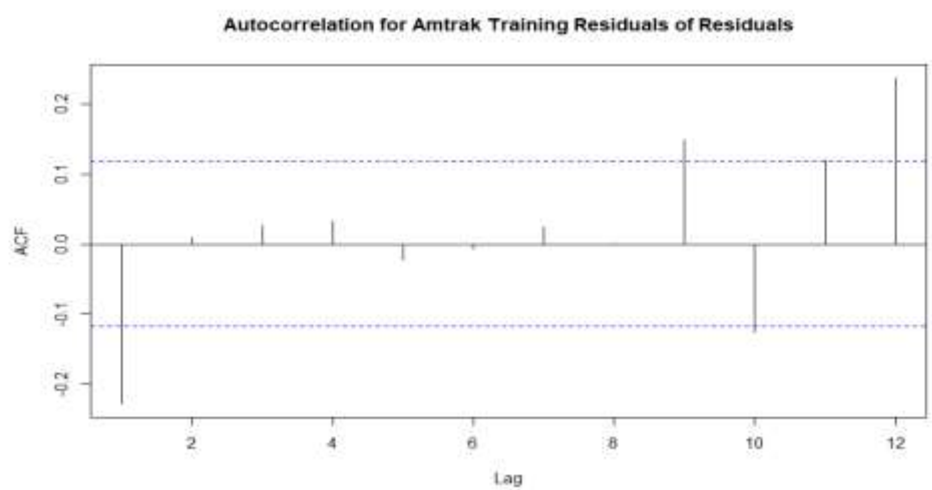
	Ridership	Regression	Residuals	AR.Model	AR.Model.Residuals
1	1708.917	1305.961	402.956	202.992	199.963
2	1620.586	1268.400	352.186	347.937	4.248
3	1972.715	1629.044	343.671	304.313	39.357
4	1811.665	1623.622	188.043	296.997	-108.954
5	1974.964	1672.966	301.998	163.274	138.724
6	1862.356	1665.571	196.785	261.190	-64.404
7	1939.860	1768.947	170.913	170.786	0.128
8	2013.264	1761.107	252.157	148.555	103.602
9	1595.657	1454.087	141.570	218.364	-76.793
10	1724.924	1597.621	127.303	123.342	3.961
11	1675.667	1595.889	79.778	111.083	-31.305
12	1813.863	1614.268	199.595	70.247	129.348

Regression Residuals and Residuals of Residuals with AR(1) Model





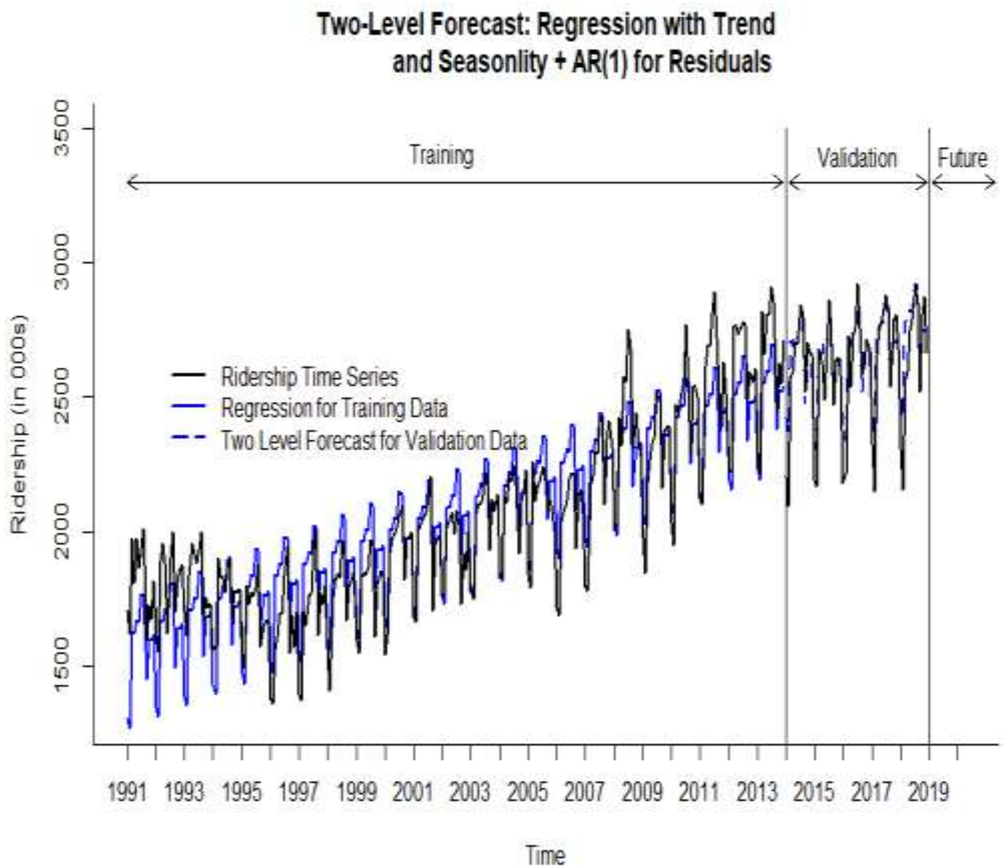




Results of Two-Level Forecasting with AR(1) Model for Residuals in Validation Period

First 12 validation records (January 2014 – December 2014) with Amtrak ridership data, regression forecast, *AR(1)* forecast for residuals, and combined forecast

	Ridership	*Regression Forecast*	*AR(1) Forecast*	*Combined Forecast*
1	2206.788	2275.614	147.847	2423.461
2	2092.819	2238.053	128.735	2366.788
3	2575.951	2598.697	112.314	2711.011
4	2592.994	2593.275	98.203	2691.478
5	2700.179	2642.618	86.079	2728.698
6	2695.954	2635.223	75.662	2710.885
7	2844.949	2738.599	66.710	2805.310
8	2802.873	2730.760	59.019	2789.779
9	2519.583	2423.740	52.410	2476.149
10	2702.607	2567.274	46.731	2614.005
11	2667.575	2565.542	41.852	2607.393
12	2656.185	2583.921	37.659	2621.580



Performance Measures of Two-Level Model vs. Regression Model in Validation Period

Two-Level Model: Linear Trend and Seasonality + AR(1) for Residuals

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-45.365	106.92	81.199	-2.004	3.301	0.428	0.47

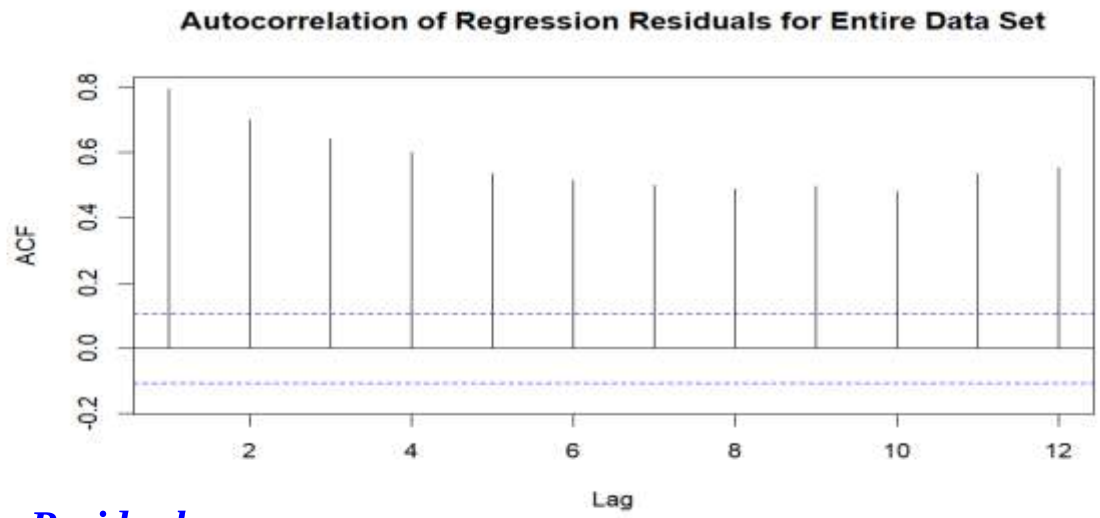
Linear Trend and Seasonality Model

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-17.224	95.541	76.238	-0.892	3.04	0.42	0.424

Forecasting for Future Periods

- **Before attempting to forecast future values of the series, *the training and validation periods must be recombined into one (entire) time series dataset***
- **The chosen model or models, e.g., *regression with linear trend and seasonality* and *two-level model (regression with linear trend and seasonality + AR(1) model for residuals)*, *need to be rerun on the entire (complete) dataset***
- ***Measure models performance vs. original data* using accuracy measure like *MAPE*, *RMSE* and others**
- **The best forecasting model, based on the entire dataset, can be now used for forecasting future periods**

Autocorrelation of Regression Model's Residuals and AR(1) Model of Residuals for Entire Data Set



AR(1) Model for Residuals

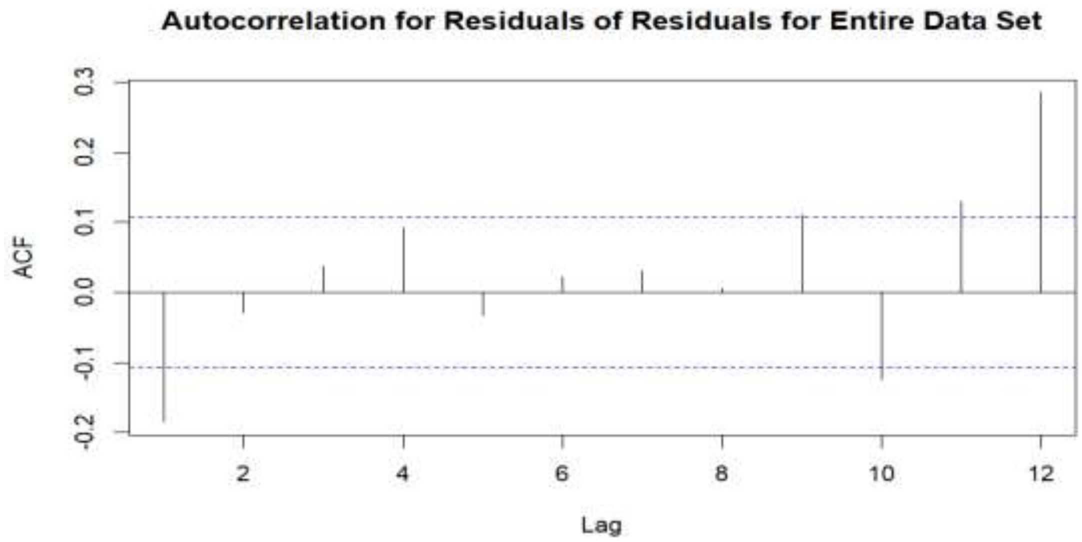
Series: lin.season$residuals
ARIMA(1,0,0) with non-zero mean

Coefficients:

	ar1	mean
	0.8162	4.3953
s.e.	0.0326	22.3605

sigma^2 = 5851: log likelihood = -1933.62
AIC=3873.23 AICc=3873.3 BIC=3884.68

$e_t = 4.395 + 0.816 e_{t-1}$

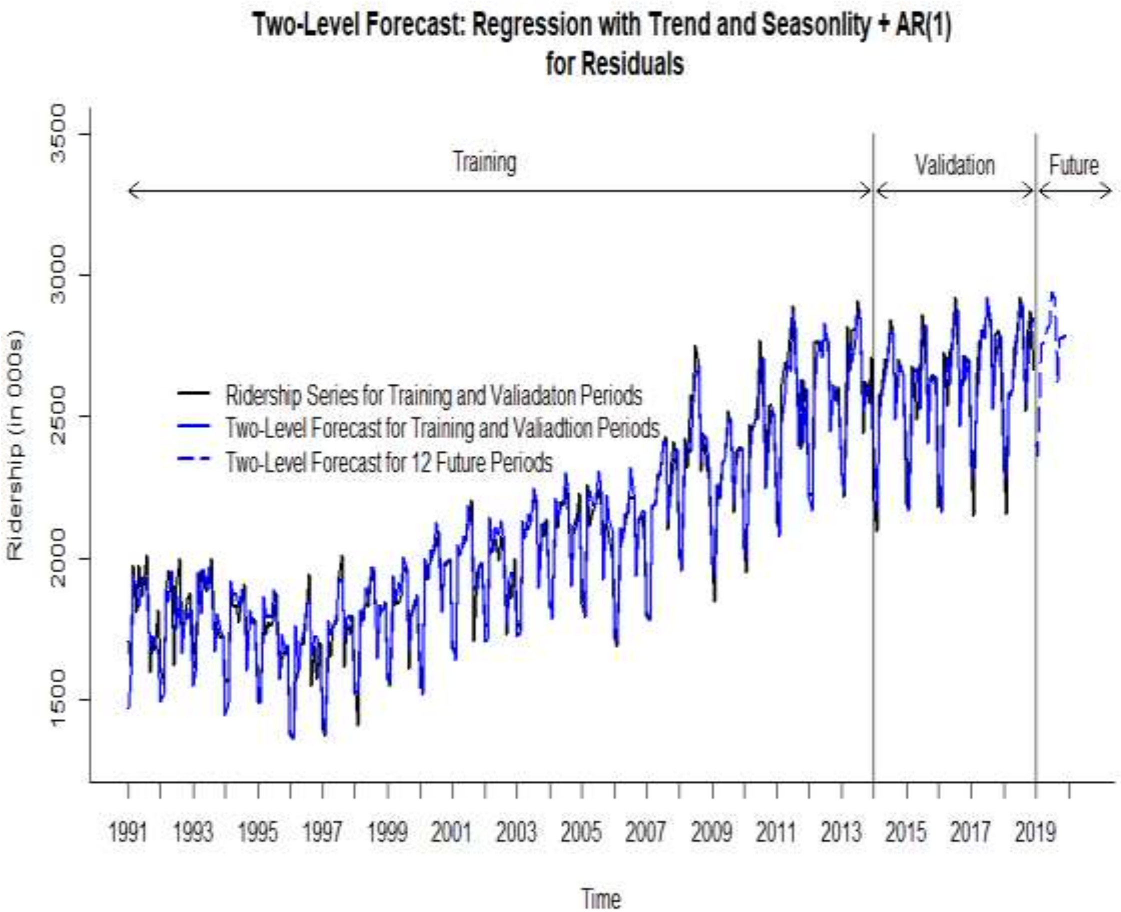


Dr. Z. Radovilsky

Results of Two-Level Forecasting in 12 Futures Periods

Forecast for 12 future periods using regression model with linear trend and seasonality, AR(1) for residuals, and combined forecast

	Regression Forecast	***AR(1) Forecast***	***Combined Forecast***
1	2456.301	-58.671	2397.630
2	2409.923	-47.078	2362.845
3	2792.287	-37.616	2754.671
4	2784.801	-29.894	2754.908
5	2836.938	-23.591	2813.348
6	2841.993	-18.446	2823.547
7	2951.144	-14.247	2936.897
8	2926.603	-10.821	2915.783
9	2628.541	-8.024	2620.517
10	2779.037	-5.741	2773.297
11	2783.325	-3.878	2779.447
12	2782.436	-2.357	2780.079



Performance Measures of Two-Level Model with AR(1) vs. Regression and Seasonal Naïve Models

Two-Level Model: Linear Trend and Seasonality + AR(1) for Residuals

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-1.512	76.267	57.927	-0.193	2.742	-0.185	0.384

Linear Trend and Seasonality Model

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	0	128.744	103.733	-0.33	5.103	0.793	0.693

Seasonal Naïve Model

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	31.113	116.076	93.101	1.26	4.375	0.631	0.579

Evaluating Predictability

- ***Predictability***
  - **Means that time series is predictable, i.e., its historical data and patterns can be used to apply for non-trivial (non-naïve forecast) type prediction with forecasting methods**
  - **Figuring out if forecasting effort may be useful**
  - **Main question: should we go beyond naïve forecast?**
- ***Random walk***
  - **Time series in which changes from one time period to the next are random**
  - **The *random walk hypothesis* in finance means that stock market prices evolve according to a random walk (so price changes are random) and thus cannot be predicted.**
  - ***Random walk series is considered non-predictable***
- **Random walk is a special case of AR(1) model**
  - **If in AR(1) model, $Y_t = \alpha + \beta_1 Y_{t-1} + \varepsilon_t$, $\beta_1 = 1$, then the model becomes:**

$$Y_t = \alpha + Y_{t-1} + \varepsilon_t$$

  - **This model is *random walk*, the difference between period *t* and *t-1* is random**
  - **Forecast becomes a *naïve forecast + parameter α***
  - **α = *drift parameter***

Predictability Test: How to Check if Time Series is Random Walk

- ***Approach 1***
  - **Fit an *AR(1)* model to time series and test the hypothesis that the slope coefficient $\beta_1=1$ (H_o: $\beta_1=1$ vs. H_1: $\beta_1 \neq 1$)**
  - **If the H_o hypothesis is rejected (*p*-value is small, below *0.05* or *0.01* confidence), then the series is not random walk**
  - **Otherwise, if H_o cannot not be rejected, this may indicate that a series is random walk**
- ***Approach 2***
  - **Mathematically an equivalent of *approach 1***
  - **Examine differenced series: Y_2-Y_1, Y_3-Y_2, … , Y_t-Y_{t-1}**
  - **If original series is random walk, the differenced series behaves like random walk**
  - **Hence, to test whether the series is random walk:**
    - » **Compute differenced series**
    - » **Examine *ACF* for differenced series**
    - » **If *ACF* plot indicates that the autocorrelation coefficients at lags 1, 2, 3, etc. are within horizontal thresholds, that it can be inferred that time series is random walk**

Calculation of First Differences for Amtrak Ridership in Excel

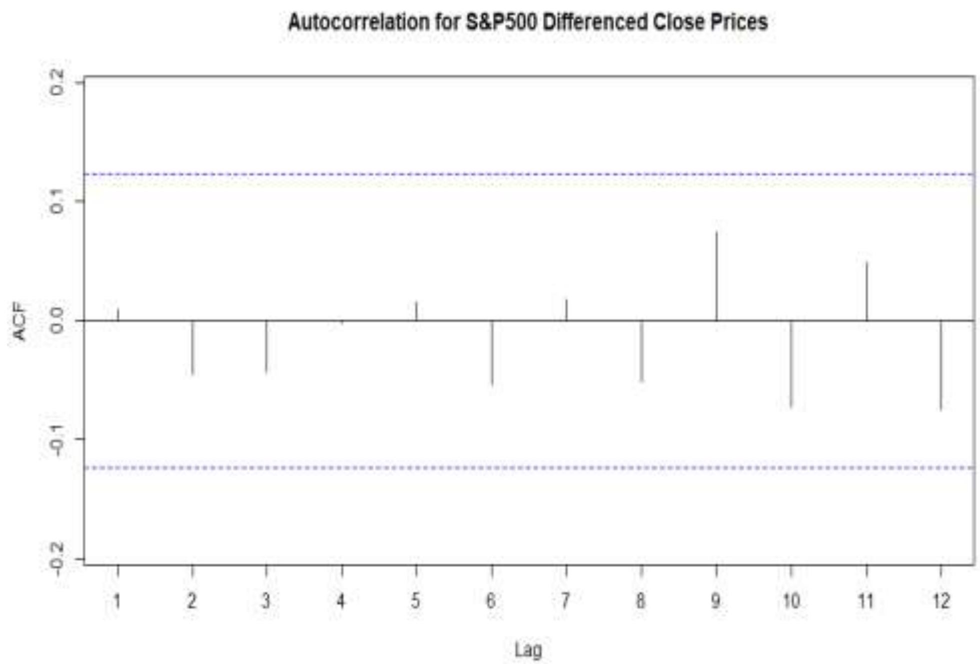
Month	*Period*	*Ridership*	*Lag 1*	*Once Differenced Ridership*
1/1/1991	1	1708.917		
2/1/1991	2	1620.586	1708.917	-88.331
3/1/1991	3	1972.715	1620.586	352.129
4/1/1991	4	1811.665	1972.715	-161.050
5/1/1991	5	1974.964	1811.665	163.299
6/1/1991	6	1862.356	1974.964	-112.608
7/1/1991	7	1939.86	1862.356	77.504
8/1/1991	8	2013.264	1939.860	73.404
9/1/1991	9	1595.657	2013.264	-417.607
10/1/1991	10	1724.924	1595.657	129.267
11/1/1991	11	1675.667	1724.924	-49.257
12/1/1991	12	1813.863	1675.667	138.196
1/1/1992	13	1614.827	1813.863	-199.036
2/1/1992	14	1557.088	1614.827	-57.739
3/1/1992	15	1891.223	1557.088	334.135
4/1/1992	16	1955.981	1891.223	64.758
5/1/1992	17	1884.714	1955.981	-71.267
6/1/1992	18	1623.042	1884.714	-261.672
7/1/1992	19	1903.309	1623.042	280.267
8/1/1992	20	1996.712	1903.309	93.403
9/1/1992	21	1703.897	1996.712	-292.815
10/1/1992	22	1810.000	1703.897	106.103
11/1/1992	23	1861.601	1810.000	51.601
12/1/1992	24	1875.122	1861.601	13.521

Predictability Test for S&P 500 Close Prices

- **Approach 1:** ***Fit AR(1) model into S&P 500 Close Prices data***

```
Series: close.price.ts
ARIMA(1,0,0) with non-zero mean

Coefficients:
         ar1       mean
      0.9843  4176.9914
s.e.  0.0115   207.2107

Apply z-test to test the null hypothesis that beta
coefficient of AR(1) is equal to 1.
ar1 ← 0.9843
s.e. ← 0.0115
null_mean ← 1
alpha ← 0.05
z.stat ←(ar1-null_mean)/s.e.
z.stat = -1.365217
p.value ← pnorm(z.stat)
p.value = 0.08609237
if (p.value < alpha) {
   "Reject null hypothesis"}
   else {
   "Accept null hypothesis"}
"Accept null hypothesis"
```

- **Approach 2:** ***Apply ACF for differenced Close Prices data***



- **Conclusion:** ***S&P 500 Close Prices in 2022 data may be random walk, can be hard to predict***

Predictability Test for Amtrak Ridership Data

- **Approach 1:** ***Fit AR(1) model into Amtrak Ridership data***

Series: ridership.ts
ARIMA(1,0,0) with non-zero mean

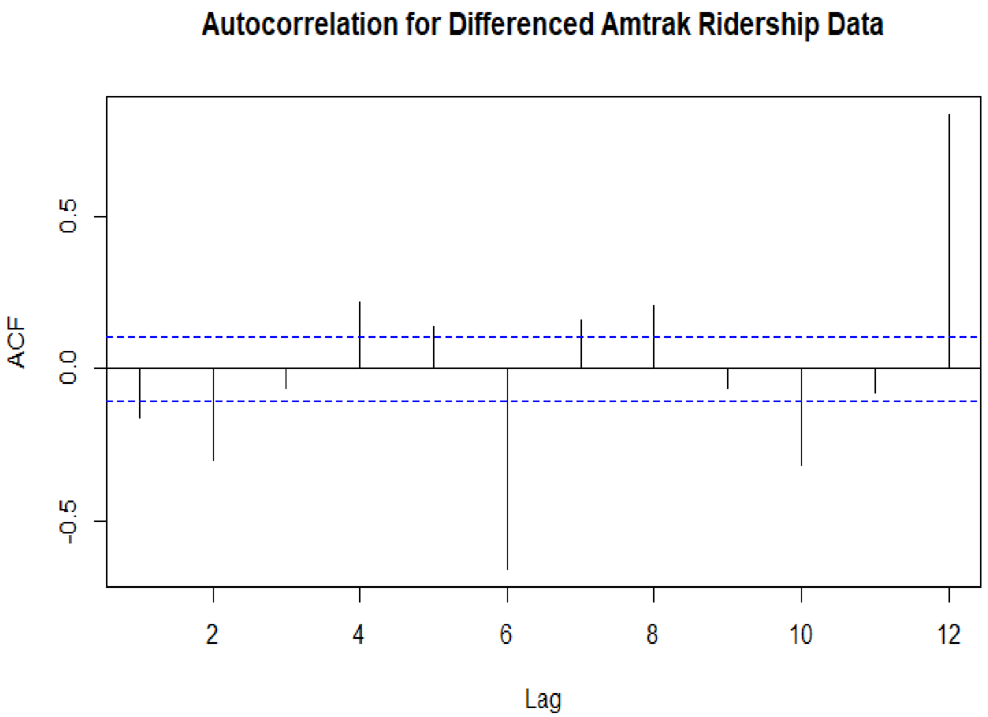
Coefficients:

	ar1	intercept
	0.8826	2147.3592
s.e.	0.0256	84.5245

Apply z-test to test the null hypothesis that beta coefficient of AR(1) is equal to 1.

ar1 ← 0.8826
s.e. ← 0.0256
null_mean ← 1
alpha ← 0.05
z.stat ← (ar1-null_mean)/s.e.
z.stat = -4.585937
p.value ← pnorm(z.stat)
p.value = 2.259769e-06
if (p.value < alpha) {
  "Reject null hypothesis"}
  else {
  "Accept null hypothesis"}
"Reject null hypothesis"

- **Approach 2:** ***Apply ACF for differenced Amtrak Ridership data***



- **Conclusion:** ***Amtrak Ridership data is not random walk, can be predicted***